

Question number	Answer	Additional guidance	Mark
9(b)	An explanation that includes three of the following points: • (epigenetic modification) prevents gene expression (1) • by {<u>DNA</u> methylation / <u>histone</u> modification} (1) • therefore transcription and (then) translation cannot occur (if gene is silenced) (1)	ACCEPT genes are {silenced / switched off / deactivated ACCEPT other examples of epigenetic modifications e.g. repressor proteins, histone methylation, histone deacetylation DO NOT ACCEPT post-transcriptional modification ACCEPT description ACCEPT {RNA polymerase / transcription factors} prevented from binding to the DNA	(3)

Question number	Answer	Additional guidance	Mark
9(c)	An explanation that includes five of the following points: (development of) cancer will stimulate a secondary immune response (1) AND ANY FOUR FROM: - because an immune response (to the envelope proteins / by the vaccine) was stimulated (1) - memory cells produced (1) - antibodies will be produced by plasma cells (1) - that will opsonise the cancer cells / so that macrophages can {engulf / destroy} them (faster) / antibodies will be produced faster (1) - T killer cells will release {perforins / chemicals / enzymes} destroying cancer cells (sooner) (1)	ACCEPT description e.g. <u>memory cells</u> produce a faster response DO NOT ACCEPT infection with cancer cells ACCEPT a description of macrophages as APCs to T helper cells active immunity / humoral response / cell mediated response DO NOT ACCEPT by B cells ACCEPT agglutination DO NOT ACCEPT antibodies destroy cancer cells	(5)

